



14. PCA AND INFORMATION LOSS



Computer Vision – Dimensionality Reduction



(a) CONCEPT EXPLANATION

Q. (a) Analyze the trade-off between dimensionality reduction and information loss, and justify optimal component selection.

Principal Component Analysis (PCA) projects high-dimensional data onto a lower-dimensional subspace by retaining the top- k eigenvectors (principal components) with the largest variances.

Trade-off:

- More reduction (smaller k) → Lower computation and storage, but higher information loss.
- Less reduction (larger k) → More information preserved, but less dimensionality reduction.

Information Loss

Information loss occurs because discarded components contain variance (details) that may represent important features, textures, or discriminative patterns.

Justifying Optimal Component Selection

Choose the smallest k that preserves most of the important information while achieving desired reduction.

Effect of Aggressive PCA (Information Loss)

Original Image (High Dimensions)	Aggressive PCA (Too Few Components)	Sufficient PCA (Optimal Components)
• All details preserved • High computational cost • High dimensionality	• Important details lost • Blurred edges, textures • Poor reconstruction	• Most details preserved • Good reconstruction • Reduced dimensionality

How PCA Works



Key Takeaway

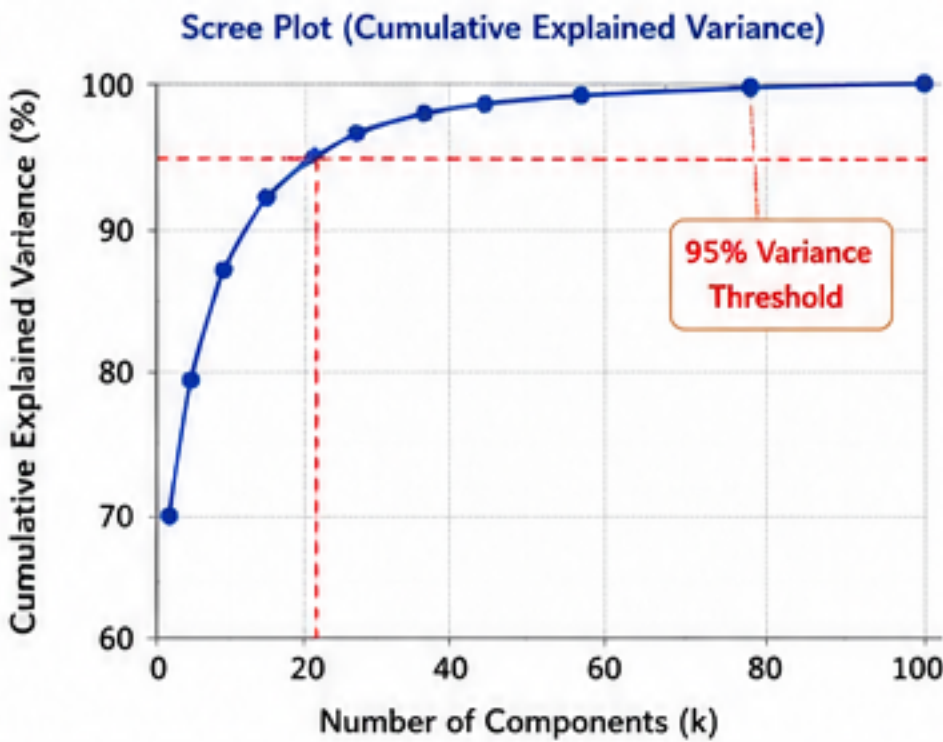
Aggressive PCA reduces dimensionality but may remove important information. Optimal PCA keeps enough components to retain maximum useful information with minimum redundancy.



(b) NUMERICAL EXAMPLE

Q. (b) Given a dataset with 100-dimensional features. The cumulative explained variance ratio (EVR) for the principal components is as follows. Analyze and select the optimal number of components.

No. of Components (k)	Cumulative Explained Variance Ratio (%)	Information Retained (%)	Information Lost (%)
5	70.2	70.2	29.8
10	85.4	85.4	14.6
15	92.1	92.1	7.9
20	95.8	95.8	4.2
30	98.2	98.2	1.8
40	99.1	99.1	0.9
50	99.6	99.6	0.4
60	99.8	99.8	0.2
80	99.95	99.95	0.05
100	100	100	0



What does the result mean?

- First 5 components retain 70.2% information (too much loss).
- 20 components retain 95.8% information (good balance).
- Beyond 30 components, gain in information is marginal (<3%).

Conclusion:

Selecting $k = 20$ components is optimal as it retains 95.8% of the information while reducing dimensionality by 80% (from 100D to 20D). This achieves a good trade-off between information preservation and computational efficiency.



(c) VISUALIZATION

Original Image (High Dimensions)	$k = 5$ Components (70.2% Variance)	$k = 10$ Components (85.4% Variance)	$k = 20$ Components (95.8% Variance)	$k = 50$ Components (99.6% Variance)	$k = 100$ Components (100% Variance)
	High Information Loss	Some Details Lost	Good Reconstruction	Very Close to Original	No Information Loss



Observation: As the number of components increases, reconstruction quality improves and information loss decreases. Choosing too few components leads to loss of important features and poor performance.

